6.1. OS Overview

- OS
  - What That System Operation (OS)?
    - System Operations (OS) are computer programs that manage source Power device hardware and devices soft .
    - OS functions involve management memory , allocation source Power system , control I/O devices , and management network as well as file system .
    - From perspective user , OS provides services and interfaces interaction .
    - For programmers , OS is interface main to log in.
    - For designers , the OS connects various modules and units for implement different functions . _
    - By overall , the OS is device soft system very significant basis .
    - After a number of decade development , computer OS has be one _ system device soft largest and most complex in the world.
  - Structure System Computer
    1. Hardware
       a. Hosts
          - Central Processing Unit (CPU)
            - Arithmetic and Logic Unit
            - Controllers
          - Main (internal) memory
            - Random Access Memory (RAM)
            - Read – Only Memory (ROM)
       b. Peripherals
          - External (Auxiliary) Memory (disk and CD - ROM)
          - Input devices (keyboard, mouse, etc.)
          - Output devices (monitor, printer, etc.)
    2. Software
       a. System Software
          - OS (DOS, Windows, etc.)
          - Programming language processor
          - Development tools, network software, and program diagnosis
       b. Application Software
          - Software packages and network suites
  - OS Types
    1. System Batch Processing Operations
       - Single-job: Program loaded to in memory and executed individually .

- Multi-job: Multiple programs are loaded to in memory and executed in a way simultaneously .

2. System Time-sharing operations

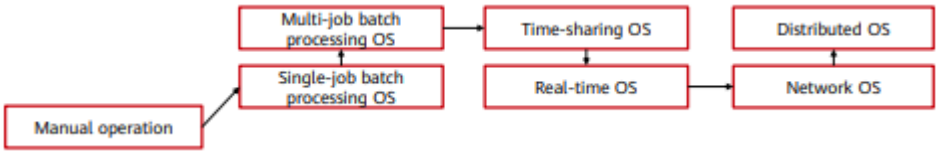
   System operation that is alternate execute every program in between user in a time slot short . Every user can interact with computer via terminal.

3. System Real- time Operation

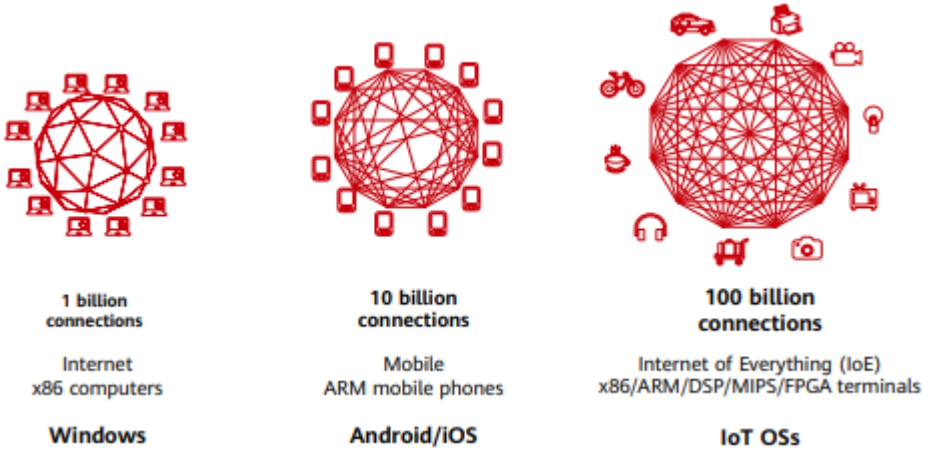
   System implementing operations _ function certain in framework specified time . _ System real-time operations are shared into soft real-time and hard real-time.

4. System Operation Network and Distributed

   Share various source Power in networking and communications between computer . Difference between system operation distributed and systems operation network is that in the first , some computer Work The same For finish the same task .



➢ From Internet to Mobile to IoE



- IoT OS

➢ Condition System IoT Operations

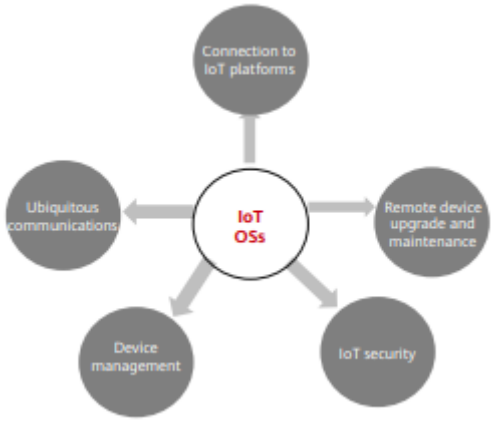
- along with spread IoT applications , consumers has become object IoT services . New IoT service own significant difference _ from services that are traditional dedicated . Operation and maintenance (O&M), architecture networking , and data processing all in all different . System IoT operations necessary develop from being separate and closed become more open and connected .

- For become open and connected , System IoT operations must can adapt with various terminal type .



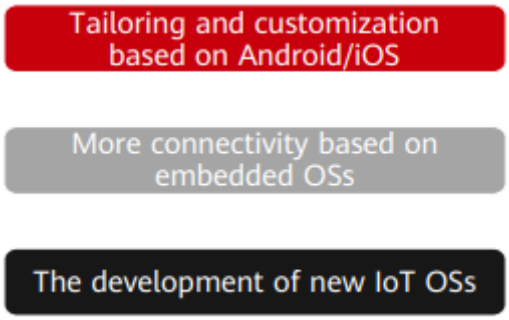
➢ System Technical Capabilities IoT Operations

For provide functionality full from IoT applications , Systems IoT operations must own ability technical following :



➢ Three Ways of Development System IoT Operations

Because of the system existing operations _ There is No can fulfil condition IoT applications , Systems IoT operations frequently need things following :



6.2 Huawei LiteOS Overview

- There are 4 features in Huawei LiteOS , namely :

- *High Real Time Performance and High Stability*
  Where Huawei can be used in industries that need it response Real *Time* more stable and accurate .
- *Ultra-Compact Kernel*
  Basic core This can customized but No more from 10kb
- *Low Power Consumption*
  Can reach Microamperes.
- Static tailoring

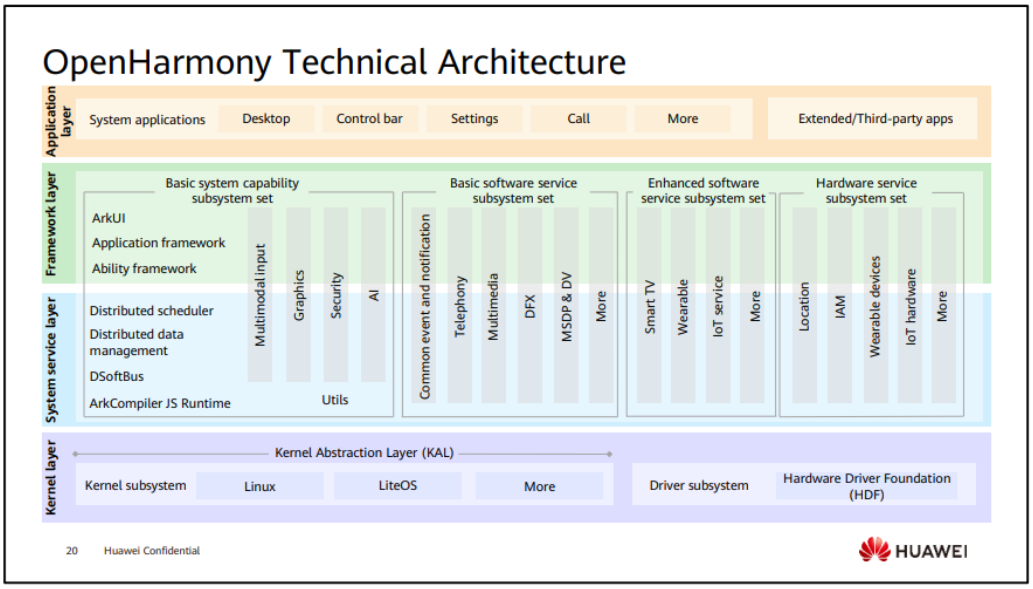
- Huawei LiteOS Technical Architecture
  A. Industry Applications
- Components
  1. OTA differential upgrade
  2. GUI frameworks
  3. AI Framework
  4. Interconnection framework
  5. Sensing framework
- Application interface layer
  1. Libc / Libm /POSIX
  2. CMSIS Adaptation Interface
  3. Protocol Stack
- File System
  1. Vfs
  2. Littlefs
  3. Fatfs
  4. Devfs
  5. Kifs
  6. Ramfs
  7. Spiffs
- Huawei LiteOS Kernels ; Formal verification
- Hardware Abstraction layer
- CPU Architecture
  1. ARM-M
  2. ARM-A
  3. ARM-64
  4. RISC-V
  B. Huawei LiteOS Kernel
- Enhanced Kernel
  1. C++
  2. Low Power Consumption (Run-Stop, Tickless)
  3. Debugging (CPU usage, Trace event tracking, LMS, Shell)
- Basic Kernel
  1. IPC (Sem, Queue, Mutex, Event)
  2. Swtmr timer software
  3. SMP mode (Spinlock)
  4. Memory Management
  5. Task Management
  6. Hardware (Interrupt Management, Exception Management, System clock tick)
  C. Huawei LiteOS Technical Features

- Low Power Consumption Framework
- OpenCPU Architecture
- Security design
- Device-Cloud Interconnection component
- Software over-the-air (SOTA) upgrades

6.3 Open Harmony

OpenHarmony is project source open which is incubated and managed by the OpenAtom Foundation , aims For build framework system operation distributed source open For device clever in all connected world scenario completely . HarmonyOS , as results commercial from OpenHarmony , offers solution For various device smart . Temporary that , HarmonyOS Connect is brand ecosystem device hard smart owned by Huawei, presents a platform for development and integration product in ecosystem device hard the .

OpenHarmony designed with architecture layered , consisting from kernel layer , layer service system , layer framework , and layers application from lower to on . Function system expanded in a way gradually , from system to subsystems , and more carry on to component .



Open Harmony Technical Features

- HarmonyOS is system unified operation _ For flexible deployment : penHarmony _ possible scalability source Power hard through design based components and scale small . This can applied in accordance request For various devices , incl ARM, RISC-V, and x86 architectures , as well provides a starting RAM volume from hundreds of KB to GB.
- Collaboration device loud and share source Power supported by HarmonyOS , on which device it is running can connected on layers system as a Super Device, it is possible share ability and provide seamless experience . _

- OpenHarmony also provides framework Work applications , capabilities , and interfaces user (UI). With framework Work here , developer can develop application once , and then with flexibility apply it in various ways different devices . _ Interface Programming Consistent application (API). ensure compatibility operational applications in all device .

Distributed Features OpenHarmony

- Distributed soft bus (DSoftBus) is base unite For connection without obstacle between device . It's empowering OpenHarmony with ability communication distributed For with fast discover and connect devices , as well transmit data online efficient .
- Distributed data management utilise DSoftBus For Manage application and user data spread across various sites device .
- Scheduling task distributed based on features technical like DSoftBus , distributed data management , and profiles distributed . It supports distance startup distant , invocation distance remote , binding / unbinding , and migration applications throughout _ device .
- Virtualization device distributed possible convergence source Power cross device , management devices , and data processing so virtual devices can works as expansion smartphone capabilities for form a Super Device.

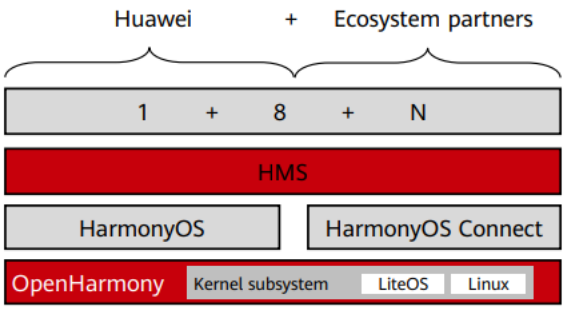
Open Harmony Types

- **Mini System:** For equipped device _ with the MCU, like 32-bit Arm Cortex-M and RISC-V processors , as well at least 128 KB of memory , system This provide various protocol network light , skeletal graphics , and variety component read and write with IoT buses. Products that can applied covers module connections , sensors, devices that can used , and various device clever For House . It supports LiteOS -M kernel .
- **Small Systems :** For equipped device _ with processor applications , such as Arm Cortex-A, and at least 1 MB of system memory This provide more security _ OK , framework graphic standard , as well ability video encoding and decoding . Products that can applied covers product House clever like IP camera , camera hole keys , and routers, as well product journey clever like event data recorder (EDR). It supports LiteOS -A and Linux kernels .
- **Standard System:** For equipped device _ with processor applications , such as Arm Cortex-A, and at least 128 MB of system memory This provide framework application complete which supports improved interaction , 3D GPU , composer device hard , various components , and various type animation . Products that can applied including appearance refrigerator class on . It supports Linux kernel.

Development Boards and Chips for OpenHarmony

System Type	Board Model	Chip Model	Application Scenario
Standard system	HH-SCDAYU200	RK3568	Entertainment, easy travel, and smart home devices, such as kitchen hoods, ovens, and treadmills.
Small system	Taurus AI Camera	Hi3516DV300	Devices with screens, such as refrigerators and head units.
Small system	BearPi-HM Micro	STM32MP157	Smart home central control panel.
Mini system	Pegasus Wi-Fi IoT	Hi3861	Smart home devices such as white goods and small home appliances.
Mini system	BearPi-HM Nano	Hi3861	Connection devices like smart street lamps, smart logistics sensors, and human body infrared sensors.

Huawei IoT OS Ecosystem



Ecosystem System Huawei's Internet of Things (IoT) operations include Huawei and partners ecosystem . HarmonyOS mainly used For Huawei products . Ecosystem partners can join with "1+8+N" IoT ecosystem for all scenario through HarmonyOS Connect.